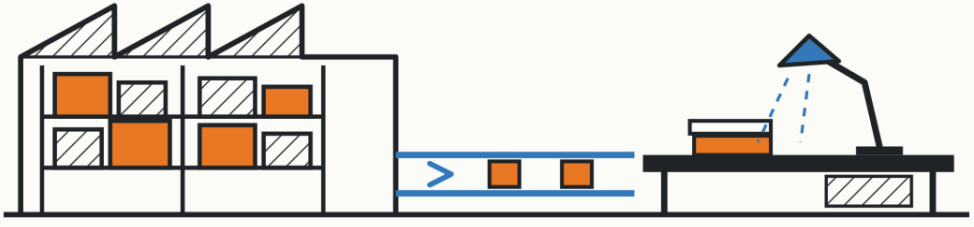
Transfers alone miss the target

One layer · one device · FP32 results
Assumed 8 TB/s for reads + writes



Target 0.200 ms

Transfer floor ≥0.243 ms



Calculated: 1.946 GB ÷ 8 TB/s.
Transfers alone exceed this target.
Compute and other traffic excluded.

Scratch becomes traffic

Attention backward · one sequence,
one layer · one device · FP32 partials



1,056 MiB produced

≤128 MiB retained on chip

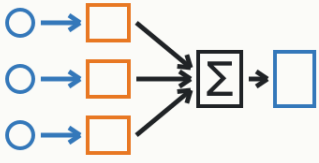
≥1,856 MiB out + back

0 1,024 2,048 MiB

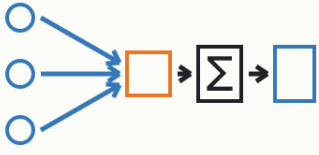
All production, then reduction.
≥928 MiB out + ≥928 MiB reread.
No compression or recomputation.

Compare the complete path

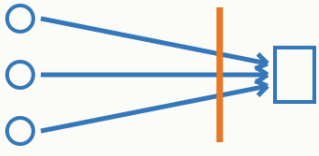
Store all partials
Separate producer outputs.
More scratch + traffic.



Combine smaller batches
Less live scratch.
More coordination.



Accumulate directly
Less staging.
Check synchronization.



Each design needs correctness checks.
Sum order can change rounding.